Week 13

Spring 2027 · Section 017 · MW 7:30–8:50 PM · RRC Room 8210

Course Level Objectives:

- Examine, analyze, and compare various sampling distributions for both discrete and continuous random variables. (CLO 5)
- Describe and compute confidence intervals. (CLO 6)
- Perform hypothesis testing using statistical methods. (CLO 8)

Module Objectives	Instructional Materials	Learning Activities	Assessments	Tools
1. Explain why inference for a mean uses the t distribution with df = n − 1. (CLO 5) **2.** Check the conditions for t procedures. (CLO 5) **3.** Construct and interpret a t confidence interval for a population mean. (CLO 6) **4.** Perform a one-sample t test and state the conclusion in context. (CLO 8)	e-Text: 6.2-CI Confidence Interval for a Mean e-Text: 6.2-D Distribution of a Mean e-Text: 6.2-HT Hypothesis Test for a Mean Interactive Lecture: 6.2 Inference for a Mean Slides: 6.2 Inference for a Mean StatKey – Theoretical Distributions (t)	Participation: Inference for a Mean – gribbles CI and Chips Ahoy test worked in groups Your Turn: Testing a Mean (juice bottles underfilled?)	Online Homework: 6.2 Inference for a Mean (due Sun Apr 25, 11:59 PM) Take-Home Quiz 9 (due Mon May 3, 11:59 PM) – covers 6.2, 6.4, 6.5 (all objectives) Test 3 (Fri Apr 30 – Sun May 2, proctored) – covers 4.4–6.5 (all objectives) Test 3 Study Guide (all objectives) Practice Test 3 (optional; due Thu Apr 29) (all objectives)	WileyPLUS StatKey Blackboard Respondus
5. Compute the standard error for a difference in two means. (CLO 5) **6.** Find the conservative degrees of freedom for a two-sample t procedure. (CLO 5) **7.** Check the conditions for two-sample t procedures. (CLO 5) **8.** Construct and interpret a confidence interval for $\mu_1 - \mu_2$. (CLO 6) **9.** Explain how the order of the groups affects the sign of the difference. (CLO 6) **10.** Perform a two-sample t test, choosing the tail from the research question. (CLO 8)	e-Text: 6.4-CI Confidence Interval for a Difference in Means e-Text: 6.4-D Distribution of a Difference in Means e-Text: 6.4-HT Hypothesis Test for a Difference in Means Interactive Lecture: 6.4 Inference for a Difference in Means Slides: 6.4 Inference for a Difference in Means	Participation: Comparing Two Means – video games and GPA CI; Pygmalion-effect test Your Turn: Exercise and Sleep (two-sample t test and CI)	Online Homework: 6.4 Inference for a Difference in Means (due Sun Apr 25, 11:59 PM) Take-Home Quiz 9 (due Mon May 3, 11:59 PM) – covers 6.2, 6.4, 6.5 (all objectives) Test 3 (Fri Apr 30 – Sun May 2, proctored) – covers 4.4–6.5 (all objectives) Test 3 Study Guide (all objectives) Practice Test 3 (optional; due Thu Apr 29) (all objectives)	WileyPLUS StatKey Blackboard Respondus